

Living Alignment

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Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means and how to reach it remains poor. Here we look to the science of life for guidance. Across scales, from genomes to ecosystems, we observe that notions of health and flourishing consistently map to the ways in which those systems limit fixation on any particular mode of interaction. Fixation is limited by barriers that contextualize modes of interaction, leading to robust modularity and open-ended invention of new structure. With these principles in mind, we turn to the alignment problem. We suggest that to be aligned in a meaningful sense, AI and human-AI systems must include evolving barriers at all scales.

Keywords: AI alignment | AI safety | evolutionary biology | open-endedness | value specification | sociotechnical alignment | human-AI systems

‘Defying definition—a word that means “to fix or mark the limits of”—living cells move and expand incessantly.’

Lynn Margulis

‘This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.’

Stuart Kauffman

In this Perspective, we draw on the science of living systems to better understand how to design and relate to AI in ways that lead to futures full of flourishing and health. We point to a principle of life which has been noted across a variety of systems at different scales. The principle is that living systems are healthy and flourishing when they avoid overly fixating on, or reducing to, particular patterns of interaction. Consistently across types of system, fixation is limited by semi-permeable barriers. Barriers translate shorter-sighted optimization pressure into productive effort, contextualizing individual entities as part of larger structures. Barriers yield deepened individuality, modularity, and open-ended discovery. In Section 1, we work through examples of these mechanics at scales from cells to societies.

Section 2 applies the same principles to the AI alignment problem. There is broad agreement that the aim of alignment—in its broadest sense—is futures full of health and flourishing. From the pattern identified in Section 1, it follows that *alignment means avoiding fixation on particular modes of interaction*. This simple lemma has a surprising consequence: any alignment method, if given too much weight, is

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necessarily *misaligned*. Instead, the flourishing of life in the future depends on an evolving set of barriers that continue to limit the influence of any particular approach or concept, creating a generative web of partially competing, partially collaborating perspectives at all scales. Ongoing invention is intrinsic to alignment.

Example	Too much influence	Barrier	Generativity
Symbiogenesis	Archea and protobacteria never differentiate	Reproductive isolation between archea and protobacteria	Invention of the eukaryotic cell
Sexual reproduction	Linkage disequilibrium, Muller’s ratchet	Sexual reproduction, out-crossing	Each gene explores a solution that can contribute to many genomes, creating combinatorial possibilities
Brains	Hypersynchrony	Homeostatic plasticity, recurrent inhibition, etc	Behavioral flexibility
Pavlovian approach	Persisting in approach even when it doesn’t work	Arbitration, cognitive control	Exploring, building a richer model
Frames	Inflexible beliefs, obsession, functional fixedness	Transition to perspective-as-object	New, more inclusive perspectives
Group problem solving	One perspective dominates, group performs worse	Structured communication, skepticism	Combining unique ideas create new inventions (eg printing press)
Institutions	Theft, monopoly, resource overuse, concentration of power	Laws, regulations, courts, etc	Individual motives become productive parts of systems
Ecosystems	Collapse of biodiversity	Predation, competition, etc	Niche creation

Table 1: Too much influence.

1 Health in living systems

The thesis of Section 1 is that health and flourishing are related to limiting fixation on particular modes of interaction. Healthy living systems instead sustain a buoyant internal structure that both admits many kinds of description and also changes over time. Conversely, systems are unhealthy when some mode of interaction gains excess traction or permanence at the expense of broader diversity, flexibility and dynamism. For example, pursuit of a single drive, such as food or sex, becomes pathological for an organism when not balanced with competing drives. At the level of ecosystems, monoculture is fragile and unsustainable. Other examples include cancer, hypersynchronous neural firing in a seizure, obsession with a particular thought, monopoly, or the imposition of one person’s will to restrict many others. We will use the term ‘collapse’ to refer to the transition to fixation.

Each entity in a living system behaves like it is trying to optimize for an objective. But that objective is necessarily myopic. If the optimization is too successful, the results are paradoxically harmful. *Without resistance, a drive discharges completely, and the proxy is consumed in place of the goal it stands for—a short circuit.* Therefore, living systems that are actually successful are the ones that place barriers or limits on the influence of individual entities. Some barriers are features of the non-biological world: the speed of light, rates of reaction, mountain ranges. But, interestingly, life also places limits on itself. Humans precommit to avoid their own future impulsivity, plants deploy elaborate barriers against self-fertilization, and a gene’s selfishness is held in check by the whole genome. This makes more sense when every living

system is viewed as a collection of parts that constrain one another. A world of many distinct parts, each following its own rules from its own local perspective, functions as a rich web of barriers that holds other parts in context. **In turn, that whole system is a part in another system, and is similarly bounded.**

To limit fixation, healthy living systems depend on semi-permeable barriers. Barriers permit interactions that are temporarily or conditionally stable, without letting them become rigid or dominating. **A semi-permeable barrier makes a path harder without closing it.** Cognitive control allows various drives and habits to motivate behavior conditionally; in ecosystems, species hold each other in check and create more complex roles for each other. When barriers are functioning well, they don't shut down a pattern of interaction, but instead contextualize it as part of a larger system.

Mechanistically, to hold parts in context, living systems rely on an extraordinary, many-scale network of semi-permeable barriers (Maturana and Varela, 2012; Moreno and Mossio, 2015; Simon, 1962) (Fig. ??). By limiting interactions, barriers divide living systems into parts, creating the conditions for each part to refine its own identity—like a writer developing a unique voice or a species adapting to a niche. **By partitioning a system, barriers are also what give each part a distinct existence: a part is whatever lies on one side of a barrier.** Barriers guard against fixation, breaking up stationary arrangements and **protecting** parts from being dominated by other parts or blended into homogeneity, either of which is fixation to a form admitting less nuanced description. But by allowing interactions, barriers enable the parts to participate in flexible relationships with one another, contextualizing each part within larger structures—structures that often could not have formed without the diverse parts having undergone their own unique development. The resulting larger system accommodates many partial perspectives rather than locking in a single story of what matters.

Life's history is littered with collapses at all scales, from the death of a cell to mass extinction events. When collapse is contextualized, the larger system can remain healthy: the death of a cell is catastrophic for the cell but may be healthy in the larger context of the organism. If instead a cell becomes cancerous, it loses the context of appropriate function within a tissue and recklessly executes its myopic vision, collapsing the tissue or even the whole organism. A larger system, made out of parts, functions well when the parts locally express their distinct identities but also participate in larger contexts; it functions poorly when a form runs amok without contextualization, overwhelming the potent interplay of diverse parts.

Semi-permeability is often highly specific. Barriers might block certain objects or information, constrain when interactions happen or how large an effect they may have, or gate information conditional on another factor. These details are themselves fine-tuned from the long evolutionary history of life, with each part carrying traces of the many contexts it has participated in.

Finally, healthy living systems are generative. Healthy living systems maximize creative potential at the delicate edge of evolving interplay between diverse, semi-independent entities. Such systems admit many kinds of meaningful description but are not perfectly or eternally captured by any of them. The life that exists today, now including humanity's storage and processing of information, is what traced a path through time avoiding both excess and insufficient stability (Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944). At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Canguilhem, 1966; Kauffman, 1992; Langton, 1990; Stanley et al., 2017). **Barriers form modular, non-overfit solutions. These become the engine of discovery.** By mitigating fixation, barriers underpin life's potential to continually discover new solutions and new patterns of organization.

We look at a few examples from a range of scales, **starting with genomes and working up to institutions.**

1.1 Genomes

Sex is costly. An allogamous organism must find a mate in a vast and dangerous world. In species with separate sexes, roughly half of the population does not bear offspring (Maynard Smith, 1971). And meiosis causes each parent to lose half its genetic representation in each child (Williams, 1975). Yet nearly all eukaryotes reproduce sexually (Speijer et al., 2015), and many hermaphrodites go to great lengths to prevent self-fertilization (Charlesworth et al., 2005). Why place a barrier on the path to reproduction?

In an asexual lineage, genes are permanently locked together: selection cannot act on one without

dragging the others. A deleterious mutation is inherited forever, unless the lineage is lucky enough to experience the reverse mutation (Muller, 1964), and two beneficial mutations arising in different individuals cannot be subsequently combined into a single genome (Fisher, 1930; Muller, 1932). Asexual reproduction or self-fertilization creates a high degree of informational connectedness between the genes of a genome. Alleles become overfit to a fixed genetic background.

Recombination breaks up associations among loci, allowing selection to act more independently on them (Felsenstein, 1974). By exposing alleles to varying genetic backgrounds, it favors alleles that perform well across contexts, promoting modularity and evolvability (Livnat et al., 2008, 2010; Wagner and Altenberg, 1996; Watson and Szathmáry, 2016). **The result is modularity, where each gene becomes an interchangeable unit that can contribute something meaningful to many different genetic contexts. This creates the foundation for faster discovery of new solutions.**

Yet the evolutionary value of genetic exchange depends on both the diversity available and how it is recombined. Mendelian inheritance preserves discrete variants rather than blending them away, but gene flow can homogenize allele frequencies between populations and, when sufficiently strong relative to divergent selection, overwhelm local adaptation (Lenormand, 2002). Partial geographic or reproductive isolation can therefore allow populations to accumulate and retain distinct variants; subsequent hybridization and introgression can bring these variants into new combinations that contribute to rapid speciation and adaptive radiation (Marques et al., 2019). At the genomic scale, the frequency and location of recombination determine which variants tend to travel together, although genes are not indivisible units: recombination can occur within them (Dooner and Martínez-Férez, 1997; Paigen and Petkov, 2010). Shuffling can assemble beneficial combinations and help selection remove deleterious mutations, but it can also disrupt combinations that work well together, so more exchange is not invariably advantageous (Otto, 2009). Models of genetic “mixability” suggest that repeated exposure to different genetic backgrounds can favor alleles that perform well across contexts, providing a possible route to modularity rather than a guarantee that every gene becomes interchangeable (Livnat et al., 2008, 2010). These findings suggest that barriers can contribute to evolutionary innovation by preserving differentiated variation while permitting its recombination; whether such arrangements suffice for open-ended evolution remains an open question.

1.2 Goals

Animals have many innate drives: for nutrition, osmotic balance, temperature, reproduction, avoidance of pain. Each evolved as a proxy for fitness, but each is imperfect, so fixation on any one paradoxically reduces fitness rather than increasing it (John et al., 2023; Kurth-Nelson et al., 2024; McFarland and Sibly, 1975). Maximizing caloric intake produces obesity; single-minded pursuit of sex produces relational, occupational and legal harm. Higher-order goals behave the same way. Focusing only on work goals can cause burnout, and maximizing reported revenue without regard for honesty or law can lead to imprisonment. Fixating on a particular *strategy* for achieving a goal can even defeat the goal it serves. In a classic experiment, hungry chickens were placed near a cup of food rigged to move in the same direction as the chicken at twice the speed, so that food could only be obtained by running away from it. The chickens were defeated by their own prepotent, overly simplistic strategy of running physically toward the food (Dayan et al., 2006; Hershberger, 1986).

Evolution has provided an array of barriers to limit fixation on particular drives, goals and strategies. One broad class of barrier is cognitive control (Botvinick et al., 2001; Diamond, 2013). Control is semi-permeable: it makes a drive’s expression conditional rather than nullifying it. By maintaining context, rules, and more distal goals, it biases competition among representations and responses. A drive, goal, or strategy can organize behavior when it fits the wider situation. Hunger can guide eating without causing overeating, and absorption in work can coexist with a rule to stop at bedtime (Braver, 2012; Miller and Cohen, 2001).

Each activated drive, goal, or strategy carries its own compressed picture of the world, and favors what appears to be the path toward a desired state under its partial representation (Carver and Scheier, 2001; Miller et al., 1960). Barriers make that path unavailable, breaking the symmetry of congruence. In Hershberger’s apparatus, a congruent response is to move toward the food. Barriers force a symmetry

breaking: in this case, moving away from the food. Symmetry breaking is a source of structure: stymying a shallow objective often leads to progress on a deeper one (Anderson, 1972; Duncker, 1945; Ohlsson, 1992; Rosenberg, 1969; Stokes, 2005).

1.3 Frames

Humans always look at the world from some perspective; there is no ‘view from nowhere’ (Nagel, 1986). Most real situations admit many valid perspectives, and each is only a partial description—famously, ‘all models are wrong’. But from inside a frame, the frame is invisible; it is simply reality (Berger and Luckmann, 1966; Goffman, 1974; Kuhn, 1970; Wittgenstein, 1953).

Losing the ability to move between frames is a signature of psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future (Beck et al., 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which, although recognised as incorrect and maladaptive, are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013).

An underappreciated domain of contextualization is awareness. Humans carry a lot of assumptions about the world, many of them never questioned. Some are lifelong and self-defining; some are fleeting and perceptual, like the assumption that the thing I’m touching is a keyboard. Within its own frame, each assumption has a kind of tautological truth: it seems so real that it is hard to think about it not being true, or to think about it at all. But philosophers, psychologists and contemplative practitioners have pointed out that there is sometimes a moment of stepping back, in which the assumed form becomes an object in awareness. We realize it is not an absolute truth standing alone but a form in our mind, and the thing that seemed axiomatic becomes just another content of experience (Kegan, 1982; Watts, 2011).

Awareness is an evolving system of barriers that limit fixation on any particular belief or mental form. The barriers are semi-permeable: becoming aware of a belief does not make it wrong in an absolute sense any more than it was right in an absolute sense, and it is held productively for the partial truth it contains. At the edge of not fixating exclusively onto particular forms, many partial truths are held in delicate balance, which activates a deeper sensitivity to our own livingness and to the world, [and a deeper generativity](#); subtler forms, which could have been erased by clamped fixation, instead play a role in a richer internal structure (Beisser, 1970).

In science, Kuhn (1970) argues that perspectives are always, necessarily, incomplete descriptions of the world. Anomalies inevitably arise from the juxtaposition of those incomplete descriptions against the real world. If the community *overcommits* to particular theories, science gets stuck.

But the point is not to shut down myopic perspectives. Any point of view is partial—this doesn’t mean we shouldn’t have viewpoints. Within the progress of science, temporary commitment to a paradigm is necessary. Kuhn and Popper (Popper, 1975) both argue that scientists should resist change to a degree. The point is to contextualize each myopic frame so it is not the sole and absolute determinant of behavior, [so that it responds to evidence](#). When science functions well, anomalies become crises and revolutions—[discovery](#).

1.4 Groups

Teams, companies, and civilizations achieve what no individual can (Hutchins, 1995; Woolley et al., 2010). Sometimes the reason is simply summed effort, like four people lifting an object too heavy for one. But the greatest achievements depend on the structured interactions of individuals who have developed their own distinct points of view. [Barriers support individuals developing their own perspectives, but because they are semi-permeable, people can also combine solutions.](#)

A corollary is that group performance can degrade when groups homogenize (Bernstein et al., 2018; Frey and Van de Rijt, 2021; Hong and Page, 2004; Janis, 1972; Lazer and Friedman, 2007; Lorenz et al., 2011). [Homogenization is a failure of insufficient barriers.](#) Zollman (2010) tells the remarkable story of how a single study failed to find bacteria in the stomach of peptic ulcer patients. The study was too influential:

the belief that ulcers are not caused by bacteria propagated through the field, and medicine settled on a wrong understanding for decades.

Social scientists have made an experimental case that limiting information sharing, under certain conditions, spares groups from homogenizing and boosts collective performance. For example, Bernstein et al. (2018) tasked small groups with solving traveling-salesman problems. In these problems, it’s possible to get stuck in dead-end solutions that initially appear promising. In Bernstein et al.’s experiment, groups performed better when they were not allowed to exchange information continuously. Working independently, group members pursued their own diverse solutions, some of which turned out to be superior—rather than picking up on seductive dead-ends from others.

In real-world problems, individuals bring unique, specialized expertise. When the nuclear submarine USS Scorpion vanished in 1968, the expanse of ocean where it might lie was enormous. John Craven, Chief Scientist of the U.S. Navy’s Special Projects Office, assembled mathematicians, submarine specialists and salvage operators. He elicited each expert’s separate input, without the experts consulting each other. He then combined their opinions via Bayes’ rule. The aggregated prediction fell less than 300 yards from the wreckage (Richardson and Stone, 1971; Sontag et al., 1998). It is conjectured that each person’s expertise related to a different aspect of the problem and thus contributed a complementary signal; and combining these yielded results better than any alone (Surowiecki, 2005).

Finally, aggregation is often modular and compositional. The steel punch was developed by goldsmiths to imprint many coins with the same design. Independently, farmers had spent centuries optimizing the screw press to extract oils and juices. Gutenberg drew on both technologies to construct his printing press (Johnson, 2010). A group with many ideas floating around, [each developed through the unique pressures of semi-isolation](#), is ripe for such compositional discoveries.

1.5 Institutions

Every actor in a society or an economy is myopic in some way, valuing its own welfare, discounting the future, or attached to a particular perspective. If unchecked, an individual actor’s optimization for its own agenda can be harmful: a company’s profit motive leading to suppression of competition, deception and exploitation (Bakan, 2006); desire for dominance leading to disempowering and silencing others (Sidanus and Pratto, 2001); even prosocial beliefs producing conflict and suppression when groups see differently (Haidt, 2012). These are all examples of fixation. A part of a system (the myopic actor) gains too much traction and collapses the richness, subtlety and diversity of the larger system.

Institutions—constitutions, laws, norms, courts, regulators—are barriers against domination by any particular actor’s motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). The point of institutions is not to annul the short-sighted intentions of individual actors but rather to contextualize them. When functioning well, institutions productively reroute energy: for example, a business’s profit motive, when constrained by competition and effective regulation, may fuel innovation. Since intelligent agents do not necessarily accept the barriers set on their desires, institutions must adapt as loopholes are found, forming an evolving network that, in the best case, acquires robustness as it is tested against many situations and motives.

1.6 Other examples

The themes highlighted in these examples appear across many systems. Cells depend on a membrane with conditional gates that admit certain signals and not others at the right time. The gradients that would annihilate the cell to equilibrium are instead put to work, [as resistance in a circuit turns a potential into work](#) (Alberts et al., 2022). In the theory of symbiogenesis, archaea and protobacteria evolved separately for hundreds of millions of years before one engulfed the other, plausibly because distinct lineages could discover solutions neither could have found from inside its own fitness well (Lane and Martin, 2010; Margulis and Sagan, 1995). [The eukaryotic cell was discovered by combining these solutions](#). Brains spend enormous resources keeping representations separate while letting them interact, from lateral inhibition to hippocampal pattern separation (Kurth-Nelson et al., 2023). Healthy brain dynamics live between pure synchrony and uncorrelated noise; collapse of this nuance is pathological, [as in the excess correlation of Parkinson’s](#)

disease (Beggs and Plenz, 2003; Hammond et al., 2007). [Imagination and discovery arise through structured synthesis of separated representations, as in replay.](#) In relationships, absence of [interpersonal](#) barriers blurs whose experience is whose. Excessively rigid barriers produce isolation and stagnation (Gabbard and Lester, 1995; Minuchin, 1974; Perls et al., 1951). Ecosystems flourish when predation, competition, and other constraints prevent any one population from overwhelming the wider community, famously illustrated by the reintroduction of wolves to Yellowstone (Ripple and Beschta, 2012). [When these constraints fail, as with invasive species, ecosystems collapse.](#) [Ecological relationships also create new niches and new functions.](#)

2 The alignment problem

In Section 1 we observed that living systems are healthy when they avoid fixation, instead holding the delicate, generative interplay of many partial perspectives. This interplay breaks down when any particular form—including randomness or homogeneity—gains too much traction, suppressing nuance and reducing creative potential. Semi-permeable barriers both promote the unique individuality of distinct entities and enable relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Under such conditions, systems open-endedly explore fundamentally new forms, avoiding collapse to static equilibrium or fixed compromise.

Now we look at the alignment problem through this lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of avoiding fixation—of maintaining creative sensitivity by continually contextualizing attachment to any particular form. [We propose that an aligned future has the generativity, robustness and modularity of life at all scales.](#) Entities at every scale gain individual autonomy as well as richer integration with larger systems. [These properties depend on evolving networks of semi-permeable barriers which contextualize new myopic processes as they arise.](#)

An advantage of this perspective is that it is grounded in principles that underpin life at a very general level, and so it is less bound to current conceptual frames, which might change radically as intelligence increases and human-AI systems evolve. There is a possibility that rapid advances in AI will lead to large changes in the organization of the world, potentially as significant as the emergence of multicellular life. We believe an alignment framework should generalize across such changes. Principles of health and flourishing that describe living systems of all kinds, at all levels of scale, are more likely to generalize. An aligned future rests on the continued generation of, and sensitivity to, new forms and concepts, including new values. We accept that humans, other living systems and AI are all tightly interwoven, with categories and definitions likely to shift over time (Bateson, 1972; Latour, 2012); a systems analysis is therefore useful for thinking about long-term flourishing.

2.1 Reward hacking

[Whatever fixed value function you write down is not ‘what you really want’ in a deeper sense.](#) The history of alignment tracks identifying particular fixations and mitigating them. For example, naive versions of reinforcement learning from human feedback train AI systems to maximize ‘likes’, which leads to obvious forms of gaming, such as flattering the user. We might naively think that AI should follow human instructions—but humans give short-sighted instructions that aren’t even in their own interests. Maybe AI should support a user’s long-term interests, but there’s this problem of aggregating across multiple people with different interests. Some believe that aligned AI should take into account the interests of other animals or ecosystems on Earth. And, ultimately, any fixed concept at all of what is good or beneficial for humanity or even for all of the earth is limited to the lenses we have available at a moment in time.

A foundational problem in AI safety is that an AI system might do what we say rather than what we mean: asked to increase human happiness, a superpowerful system that complies literally might lock us all up and electrically stimulate our brains. Modern systems do not rely on concise specification of a single objective—they build in many layers of detail about human preferences through post-training, prompting and safety filters, which keeps them behaving reasonably much of the time. But there is a widely recognized danger that even with our intentions expressed in great detail, the expression may still deviate from what

we really mean (Amodei et al., 2016; Bostrom, 2014; Krakovna et al., 2020).

The underlying reason is that our real values are incredibly complex. Humans have many kinds of values (Nagel, 1979). We value the short-term hedonic impact of sugar on the tongue; we also value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet. We value different things at different times, and different people have different values (Leibo et al., 2026; Rawls, 1993). Given time and resources to deliberate, we often attest to different values than when we make snap decisions (Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Nussbaum, 2001; Polanyi, 1966). And our values continue to evolve as we ourselves change and learn (Dewey, 1939). Any way we have of expressing our values at a given point in time is incomplete, like a planarian without the concepts to entertain human values.

So if a powerful system does a good enough job optimizing for any given value function, the results are harmful. Such perverse instantiation is a fixation on the myopic form of one particular way of specifying our values—a specification insensitive both to existing structure it does not capture, such as things we mean but do not say, and to things that will come into existence in the future.

A great deal of research in AI safety has investigated solutions to these problems. Consider the example of inverse methods, which address the difficulty of specifying values explicitly by learning a value function implicitly from behavior or other indirect signals. Even a complex learned value function is likely to be imperfect: it may deviate from our values in novel or nuanced situations, and it may fail to capture how our values change over time. It would clearly be misaligned to hand a frozen version of it to a powerful AI system as the final ground truth about what is good. More advanced methods therefore treat the value function as uncertain and needing continual update from human input, so that an agent unsure of what humans want becomes risk-averse and open to correction—interpreting a human’s attempt to turn it off as information about human preferences (Hadfield-Menell et al., 2016; Jeon et al., 2020). In this way, perverse instantiation might be mitigated by designing systems that want to adhere to human preferences but maintain explicit uncertainty about them (Hadfield-Menell et al., 2017; Russell, 2019; Shah et al., 2020). Such methods may also accommodate values that change over time.

Viewed through the lens of this paper, we can take away two things about such methods. First, they are powerful barriers limiting fixation on one reward function. Second, they are necessarily incomplete. Any particular method for learning values is still fixated if there are aspects of it that are constant across time as AI systems become more powerful and the world changes. Rather than being fixated on a particular value function, the system may be fixated on something more abstract: what counts as human behavior, what counts as a human, how to map behavior to values, the formalism for describing values, how to represent and update uncertainty, how value is translated into action. If any of these formalisms are fixed while the systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

2.2 No scheme is alignment

In addition to the example of reward hacking, there are many familiar cases where fixation

More generally, each method has a core limitation. In the framework of living alignment, any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore *no particular approach can achieve alignment*. An AI system could fixate on the language for describing the space that goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on. This has been studied extensively for certain categories of fixation, particularly fixation on values. The observation we add is that fixation on *any* form is misaligned; as a consequence, any particular alignment scheme, if fixed or taken too seriously, is necessarily misaligned. This is the lesson of life: healthy living systems are endlessly generative, with many scales of partially separated perspectives loosely and modularly coupled.

Importantly, this does not mean we should have no scheme. Quite the opposite: discarding schemes capriciously can be as much a loss of subtlety as any other particular form. Many existing approaches in AI safety are highly valuable, and continuing to discover new schemes and formalizations is essential to making progress.

2.3 From alignment to living alignment

It follows that it is not enough to simply ‘do what’s good for humans’ in a naive sense. That notion breaks down. Humanity and AI may both change substantially, and future worlds may contain patterns of organization that do not exist yet—perhaps some that would be incomprehensible to us at present. Things will change, including what there is to value and how it is valued. Think of the world today compared to the world a billion years ago. Any formalism or concept we have now will eventually appear as limited as a single-celled organism’s notions of ‘what is good and bad’.

Therefore, the problem of alignment, if carefully inspected, transforms into the open-ended problem of living alignment. In order to be aligned in the long run, AI must become more lifelike. Otherwise, as it has more power, that power will fixate particular forms. Living alignment means life-like systems that open-endedly discover new ways of contextualizing any myopic processes, preserving and extending the richness and generativity of life. As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable barriers to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

2.4 Is AI special?

With AI, things change faster, and there is a lot of power behind fixed forms. In the past, there have always been arms races between different patterns of organization, and sometimes one gets so far ahead that the local area is decimated. But if a version of AI gets so far ahead of everything else, there is the possibility of world-level destruction. AI systems increasingly are acting autonomously, faster than humans can follow, in ways humans cannot understand.

Why should AI be different from past technologies? We do not expect what we build today to be perfect; we expect to keep finding and fixing problems, which is what humans have always done. However, the iterating process may not continue to work as well as it has in the past. AI already has remarkable properties: rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading of much cognitive activity, and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement without being limited to a restrictive biological substrate. Compared with past technologies, these properties may make it harder to iterate on imperfect solutions (Bostrom, 2014), and any particular forms or assumptions built into AI systems could have substantial consequences.

Example: concentration of power. Humans have different interests, and AI is likely to be aligned to the interests of some more than others (Gabriel and Keeling, 2025; Sorensen et al., 2024). Advanced AI might also confer substantial power on those who control it, raising the possibility that a small number of humans use that control to dominate others—more likely as persuasive power increases (Costello et al., 2024), autonomous weapons place lethal force in few hands (Scharre, 2018), surveillance improves, and the need for human labor decreases (Susskind, 2020). In either case—AI’s unfairness or human power consolidation—extreme inequality risks collapsing society to the goals of a few individuals. Of course, a world where everyone is identical would be another form of fixation; distributing resources and power mitigates against fixation to the extent that it enables many diverse identities to thrive at multiple scales. It could also go the other way: because roughly the same AI technology is available to the poor and unskilled as to the wealthy and skilled, some have proposed that AI will have a levelling effect (Brynjolfsson et al., 2025).

Example: conceptual monoculture. Conceptual monoculture is fixation on particular beliefs, ideas, frames, values or problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: within one person, between humans, between humans and AI systems, and between organizations, fields, cultures and nations. Its collapse threatens fragility and lower performance (Page,

2008). Current AI systems draw from a conceptual manifold that is, in some ways, impoverished relative to humans (Messori and Crockett, 2024), and recent studies find that while individual AI outputs may be judged superior to human ones, they are also more homogeneous (Doshi and Hauser, 2024; Padmakumar and He, 2023). This narrow manifold might get broadcast to the whole world: at least a billion people now use AI for everything from relationship advice to industrial maintenance (Chatterji et al., 2025), yet because frontier models are difficult and expensive to produce, that usage is routed through a handful of models (Bommasani, 2021), risking global loss of diversity (Sourati et al., 2026). And because humans are influenced by AI while human outputs are training data for future models, homogenization could become recursive (Chaney et al., 2018). It is worth noting, however, that none of these studies made a best-effort attempt to use AI thoughtfully to increase rather than decrease diversity; with exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives.

2.5 Healthy for whom?

Our framework defines aligned systems as those with lifelike properties of multi-scale agentic modularity, which avoid collapse onto fixated forms. Is this ‘alignment’? Is it alignment to ‘human values’? How can you align to something if you start with the assumption that that thing is going to fundamentally change? Alignment is typically concerned with human interests. What happens if AI replaces humans with its own lifelike systems, but destroys humanity in the process? As we’ve discussed, fixation and health are always relative. Local fixation is part of larger flourishing, and of course the interests of different entities sometimes trade off against each other.

How is the flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge. Eventually, if there are descendants of humans, they will presumably be drastically different from what we look like at present. Holding things in stasis eternally is not what we want. Even though current human values are unlikely to be the ultimate endpoint of the universe, they are an important part of its tapestry and depth.

However, in the short to medium term, honoring existing life and values is where it’s at. It is difficult to overstate the amount of complexity and content in existing life, including in the relationships between organisms and in the mental lives of humans. Existing life has been through so much to get here: billions of years of parallel experiments all over Earth. Living alignment is inclusive of this diversity of evolving values. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without being overwritten or homogenized, and that these parts never-endingly join in structures that sustain their individuality while situating them as only part of a larger story. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality saturated with meaning and importance, and the diversity of human—and perhaps non-human—values is part of that context. Rather than short-circuiting into fixated modes, an aligned future continues to expand toward solutions inclusive of more and more apparent contradiction.

2.6 Going forward

The industry is already building in internal barriers. We apply safety post-training, guardrail models, red teaming and ongoing evaluation, and we try to improve our mechanistic understanding of AI systems in order to detect and correct hidden flaws (Olah et al., 2020). OpenAI now continuously monitors the activations and thinking traces of models, to detect when internal activity becomes suspicious, for example if models are attempting to circumvent safeguards (OpenAI, 2026). Fair principles have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025), and governments apply oversight to anticipate and mitigate risks to human rights and national security (European Parliament and Council of the European Union, 2024). Concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022), or by redistribution mechanisms (O’Keefe et al., 2020). OpenAI and Anthropic have paused development or release of frontier models to strengthen safety mechanisms (Anthropic, 2026; OpenAI, 2026). Some of these barriers have humans in the loop to greater or lesser degrees, like governmental oversight, mechanistic interpretability and scalable

oversight. Others are fully automated, like automated chain-of-thought monitoring, agentic red teaming and guardrail models.

Again, any fixed conceptual approach is misaligned. The essence of living alignment is transitioning to a system—which includes humans, AI, and the other species and ecosystems on Earth—that carries forward the full momentum and generativity of life, rather than fixating. As AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable barriers, because the potential modes of collapse will keep changing as the world changes. We do not know how likely it is that appropriate barriers will emerge. As in triage, there are three possibilities: myopic forms will inevitably be supercharged by progress in AI beyond the ability of anything else to bound them, resulting in disaster; or appropriate barriers will inevitably emerge to contextualize any forms of AI; or the outcome depends on the choices we make now.

The coarsest versions of fixation are obvious: paperclips, bioweapons, extreme concentration of power. What we have to be careful about are less obvious versions. A subtle fixation at first appears harmless, but the system as a whole is collapsing toward something without contextualization. The pessimistic argument is that AI is eroding barriers beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. They might facilitate rogue actors building weapons of mass destruction, or reduce the ability of democratic systems to check the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people and information systems across the globe become hyper-correlated, it is as if there were a single superorganism on Earth, which may reduce both resilience to shocks and the potential for future evolution—and past evolution always depended on running parallel experiments with diverse and separate organisms. As we reviewed with problem solving in groups, there is a simple concept in modeling studies that excessive network connectivity is prone to reaching mediocre solutions quickly and losing the diversity that sustains potential for reaching higher quality solutions over the long run (Lazer and Friedman, 2007; Zollman, 2010). Conceptual monoculture leads to an impoverished future, or a loss of drive.

The optimistic argument is that AI is creating more and more structure in the world, including the appropriate barriers to contextualize its own new abilities. We have AI tools for bounding AI: detecting bugs before others can exploit them, detecting deepfakes, scalable oversight. Appropriately-tuned AI can facilitate human learning and nurture each individual’s unique perspective and curiosity, and democratic AI could help humans with different perspectives co-exist and cooperate. Advances in technology may create new niches faster than they destroy old ones, and there is plenty of space on Earth for increasing diversity.

The good version increasingly looks like subject-to-object perspective-taking within AI and human-AI systems. At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery—transformation of the subject itself—is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life. This is internal discovery within systems that represent themselves and the world.

The melioristic argument is that because it is easy to imagine concrete paths toward both outcomes, both may be within the realm of possibility; and that as long as there is any possibility our choices are decisive, we are obliged to act as if they were (Ord, 2020). We lean toward the melioristic position. If it is right, then continual effort toward invention will be required—first from humans, and then increasingly from AI or human-AI systems—to contextualize myopic forces. Many of the ways this effort would be applied are already well studied in AI safety. Others will have to be discovered. No set of barriers will be a final answer.

The ideas in this paper come with the same proviso: we should not over-attach to them. Our aim is to

offer a conceptual lens, which is of course imperfect and temporary.

2.7 Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment.

Instead, an aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include them. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other’s sharp edges as constructive momentum.

Alignment itself is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

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